SHUAIKE SHEN

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Education

Carnegie Mellon University — Pittsburgh, PA, U.S.
Ph.D in Computational Biology — *Aug. 2025 - Present*

Zhejiang University — Hangzhou, Zhejiang, China
B.Eng in Computer Science and Technology (Honors) — *Sept. 2021 - June. 2025*

- Mixed Class, Chu Kochen Honors College
- GPA: 3.83/4.00

Research Interest

I am deeply interested in AI for science and deep generative model, especially for protein engineering, molecule design, and flow matching models. Previously, I also worked on topics related to scientific LLM, reinforcement learning and crystal structures.

Research Experience

Research Assistant — **Carnegie Mellon University**
Advisor: Prof. Olexandr Isayev — *Aug. 2025 - Present*

- Working on low energy conformer generation using diffusion models.

Research Intern — **Shanghai AI Lab**
Advisor: Dr. Biqing Qi and Dr. Yuqiang Li — *May. 2025 - Aug. 2025*

- Working on large language models for molecular spectra and developing molecular foundation models to bridge spectral data with molecular properties and structures.

Research Intern — **University of Illinois Urbana-Champaign**
Advisor: Prof. Ge Liu and Prof. Nicholas Ching Hai Wu — *June. 2024 - Mar. 2025*

- Latent Flow Matching for Protein Design: Developed a latent flow matching approach for joint protein structure–sequence design, resolving inconsistencies in prior two-stage models and achieving improved scalability and performance.
- RL for Flow Matching: Applied reinforcement learning with regularization to flow matching models, enabling diverse and stable generation while avoiding model collapse.

Research Assistant — **Zhejiang University**
Advisor: Prof. Chunhua Shen and Prof. Hao Chen — *Oct. 2022 - May. 2024*

- Contributed to *VFN*, a novel SE(3)-invariant graph encoder that overcomes the atom representation bottleneck of IPA encoders, achieving superior results in Inverse Folding and Protein Design.
- Developed *FADiff*, a protein multi-motif scaffolding model that treats motifs as rigid bodies, achieving strong performance and generalization from training on two motifs to scaffolding up to five.
- Designed a denoising pre-training framework (*DPF*) for crystals, leveraging unlabeled structural data to improve modeling of crystal structure–property relationships.
- Proposed *RiFold*, an RNA inverse folding model with autoregressive decoding to enforce correct base pairing, achieving state-of-the-art results across multiple benchmarks.
- Introduced *CEBind*, an unsupervised model for binding energy prediction based on energy conservation, providing a fast and efficient method for complex design.

Publications

* denotes equal contribution and † denotes corresponding author.

SpectrumWorld: Artificial Intelligence Foundation for Spectroscopy Aug. 2025

- Zhuo Yang*, Jiaqing Xie*, **Shuaike Shen**, Daolang Wang, Yeyun Chen, Ben Gao, Shuzhou Sun, Biqing Qi, Dongzhan Zhou, Lei Bai, Linjiang Chen, Shufei Zhang, Jun Jiang, Tianfan Fu, Yuqiang Li
- **Under review**
- Arxiv, [Link]

From Sentences to Sequences: Rethinking Languages in Biological System July. 2025

- Ke Liu*, **Shuaike Shen***, Hao Chen†
- **Under review**
- Arxiv, [Link]

Online Reward-Weighted Fine-Tuning of Flow Matching with Wasserstein Regularization Jan. 2025

- Jiajun Fan, **Shuaike Shen**, Chaoran Cheng, Yuxin Chen, Chumeng Liang, Ge Liu†
- **ICLR 2025**, [Link]

A Denoising Pre-training Framework for Accelerating Novel Material Discovery Dec. 2024

- **Shuaike Shen**, Ke Liu, Hao Chen†
- **AAAI 2025 AISI**

Training Text-to-Image Flow Models on Self-Generated Data Nov. 2024

- Jiajun Fan, **Shuaike Shen**, Chaoran Cheng, Chumeng Liang, Ge Liu†
- **Under review**

Physics-Informed Neural Networks for Unsupervised Binding Energy Prediction Aug. 2024

- Ke Liu*, **Shuaike Shen***, Hao Chen†
- **Under review**
- Manuscript, [Link]

Boost Your Crystal Model with Denoising Pre-training June. 2024

- **Shuaike Shen***, Ke Liu*, Muzhi Zhu, Hao Chen†
- **ICML 2024 AI for Science Workshop**, [Link]
- Project Page, [Link]

FADiff: Floating Anchor Diffusion Model for Multi-motif Scaffolding May. 2024

- Ke Liu*, **Shuaike Shen***, Weian Mao*, Xiaoran Jiao, Zheng Sun, Hao Chen†, Chunhua Shen†
- **ICML 2024**, [Link]
- Project Page, [Link]

De novo protein design using geometric vector field networks Oct. 2023

- Weian Mao*, Muzhi Zhu*, Zheng Sun*, **Shuaike Shen**, Lin Yuanbo Wu, Hao Chen†, Chunhua Shen†
- **ICLR 2024 spotlight**, [Link]
- Project Page, [Link]

Selected Projects

VITON: High resolution virtual try on based on generative models 2023
Scientific research projects of Chu Kochen Honors College advised by Prof. Hao Chen

MiniSQL: A database system implemented in C++ 2023
The course project of Database systems and concepts

Parallel optimization of big data pivots selection based on vectorization and OpenMP 2022
High Performance Computing 101 summer course | ACM-China IPCC 2022

Skills

Programming Skills: Python, C/C++, PyTorch, SQL

Language Skills: Mandarin (Native), English (TOEFL 104)